
Unit – 2 : Session – 10 : SLO – 1 | Process Synchronization — Match the Following

Concepts with Descriptions

1. Critical Section — C. Code that must be executed by only one process at a time
2. Semaphore — B. A shared variable that uses P() and V() operations for process synchronization
3. Mutex — D. A software or hardware lock to prevent multiple threads from accessing data
4. Race Condition — H. Condition that occurs when multiple processes access shared data concurrently
5. Busy Waiting — I. Occurs when a process continuously checks a condition in a loop
6. Deadlock — F. Undesirable situation where processes wait on each other forever
7. Entry Section — G. Part of the code before entering the critical section
8. Exit Section — J. Part of the code after leaving the critical section
9. Peterson's Solution — A. A technique for solving synchronization in two-process systems
10. Bounded Buffer Problem — E. A classic IPC problem involving producer-consumer coordination